



# Data Analysis of Estonian Theatres from 2014 to 2025

**eesti  
teatri  
agentuur**

★ *October*

2026

| Prepared to:                                                                                  |                           | Prepared by:                                                   |  |
|-----------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------|--|
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## The Team

The team was formed around a shared passion for theater culture.

## Project Overview

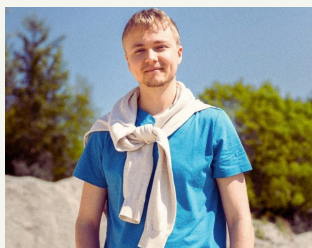
The goal is to analyze data for the Estonian Theatre Agency. The analysis focuses on Estonian theatre productions, specifically their lifespan and performance activity indicators for the period 2014–2025.

## About the organization

The Estonian Theatre Agency promotes Estonian playwriting, manages the copyrights of authors and translators, collects theatre information and statistics, and organizes a contemporary performing arts showcase festival to introduce Estonian theatre abroad.

## About Us

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### Hannes Ruusmaa

Between Tech and People,  
Project Management,  
Business Development

Every technological  
revolution starts as an  
engineering problem  
and ends up as a  
people problem.



### Vesta Väljan

Sales Manager, Account  
Manager

I am a Strategic  
Account Manager who  
speaks both the  
language of business  
and the logic of code.



### Raul Sõrmus

25+ Years in Finance &  
Purchasing, Financial  
Controller, Data Analytics

I am Senior finance  
professional with over  
25 years of expertise  
in Financial  
Management and  
Purchasing.



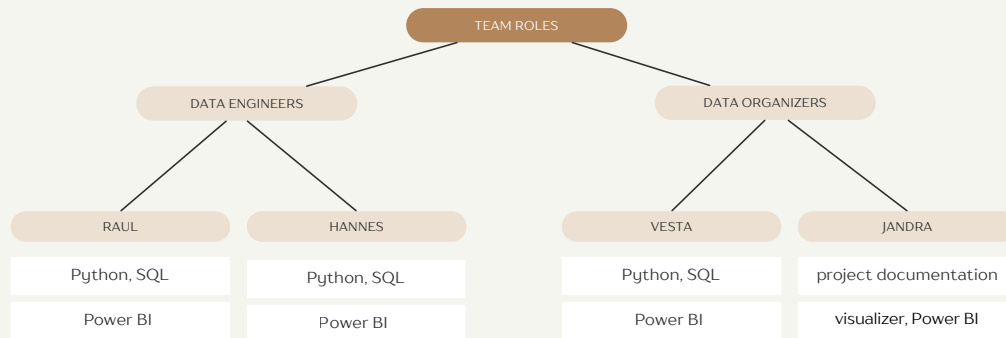
### Jandra Mölder

Project Assistant, Teacher,  
Event organizer, Project  
manager

A master's degree in  
the field of  
education, but also a  
project manager and  
developer with long-  
term experience.

# Our Team

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## Project Overview

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### Object of Analysis

The analysis focuses on Estonian theatre productions, specifically their lifespan and performance activity indicators for the period 2014–2025.

The analysis focuses on the production as an artistic whole rather than individual performances. To assess a production's lifespan and success, it utilizes aggregate data concerning performances, audiences, and ticket sales.

The analysis is based on the terminology of theatre statistics from the Estonian Theatre Agency and distinguishes:

- production – the artistic whole that is performed;
- performance – a single staging of the production;
- visitor – an audience member present at a performance, including those attending with invitations or tickets;
- ticket sold – a visitor who purchased a paid ticket.

## Data processing

| TEATER      | Omanik     | Autor /           | Autor v    | Teksti t  | Lavastuse pealkiri        | Lavastuse liit   | Žanr       | Esietendus | Uuslava | Sihtgrupp                      | Statsionaarn     | Istehtac |
|-------------|------------|-------------------|------------|-----------|---------------------------|------------------|------------|------------|---------|--------------------------------|------------------|----------|
| Piip ja Tuu | eraetendus | asutu: Dajan Ahn  | Eesti      | näidend   | Me mängime armastu        | sõnalavastus     | draama     | 06.08.2019 | 2019    | täiskasvanud: Piip ja Tuut Tea |                  | 90       |
| Piip ja Tuu | eraetendus | asutu: Diana Lee  | Eesti      | näidend   | Offline                   | sõnalavastus     | draama     | 04.10.2018 | 2018    | noored: 17 - 2                 | Piip ja Tuut Tea | 90       |
| Piip ja Tuu | eraetendus | asutu: Eduard Vil | Eesti      | dramatise | Piip ja Tuut Vilde juures | sõnalavastus     | komöödia   | 07.03.2015 | 2015    | lapsed: 6 - 11                 | Vilde muuseum    | 44       |
| Piip ja Tuu | eraetendus | asutu: Ellen Niit | / Eesti    | dramatise | Onu Ööbik                 | mitmeliigilavast | sõna, muu  | 24.05.2025 | 2025    | lapsed: 6 - 11                 | Piip ja Tuut Tea | 90       |
| Piip ja Tuu | eraetendus | asutu: Engelbert  | Saksamaa   |           | Piip ja Tuut ooperis -    | muusikalavastu   | ooper      | 29.06.2019 | 2019    | lapsed: 6 - 11                 | Harju maakond    | 0        |
| Piip ja Tuu | eraetendus | asutu: Erinevad   | Eesti      |           | Piip ja Tuut magavad j    | muusikalavastu   | kontsertla | 09.12.2017 | 2017    | lapsed: 6 - 11                 | TTÜ aula, Tallin | 0        |
| Piip ja Tuu | eraetendus | asutu: Erinevad   | Muu/erinev |           | Ai Heit Mjuuzik           | muusikalavastu   | kontsertla | 31.03.2016 | 2016    | täiskasvanud: Piip ja Tuut Tea |                  | 90       |
| Piip ja Tuu | eraetendus | asutu: Gunnar Ec  | Rootsi     |           | KASSID miisukal           | muusikalavastu   | muusikal   | 01.11.2019 | 2019    | lapsed: 6 - 11                 | Piip ja Tuut Tea | 90       |
| Piip ja Tuu | eraetendus | asutu: Haide Mäi  | Eesti      |           | Piip ja Tuut Tipp-Topp    | mitmeliigilavast | sõna, muu  | 01.04.2012 |         | lapsed: 6 - 11                 | Piip ja Tuut Tea | 90       |
| Piip ja Tuu | eraetendus | asutu: Haide Mäi  | Eesti      |           | Piip ja Tuut Kinol        | mitmeliigilavast | sõna, muu  | 15.10.2011 |         | lapsed: 6 - 11                 | Piip ja Tuut Tea | 90       |
| Piip ja Tuu | eraetendus | asutu: Haide Mäi  | Eesti      | dramatise | Piip ja Tuut loomamu      | iisõnalavastus   | komöödia   | 25.01.2014 | 2014    | lapsed: 6 - 11                 | Piip ja Tuut Tea | 90       |

Table: Performance data uncleaned Excel table Estonian Theatre Agency public data

## Description of Data Sources and Data Flow

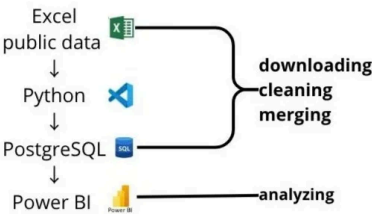
### Data source and flow

The primary dataset comes from the Estonian Theatre Agency and includes performance records from 2014–2025, supplemented by open data from the Business Register. The raw data was provided in Excel and CSV formats.

Specifically, the project utilized the following data sources:

- Estonian Theatre Agency production statistics: – data on performances, audiences, and economic indicators (ticket revenue);
- Public Estonian Theatre Agency repertoire data – information on productions and repertoire;
- Business Register – annual theatre data for the period 2019–2025;

The data flow progressed through the following steps:



Public data sources

- ↓
- Python – downloading, cleaning, linking, and transforming data
- ↓
- PostgreSQL – processed data, fact tables, dimension tables, metrics and views
- ↓
- Power BI – analysis and visualization

This approach kept the preparation and visualization stages separate, creating a reusable data pipeline that can easily be updated when new data comes in.

The raw data was processed using Python, where text fields and dates were standardized, datasets were cleaned, and matching productions were identified across various sources. These cleaned staging tables were then loaded into a PostgreSQL database.

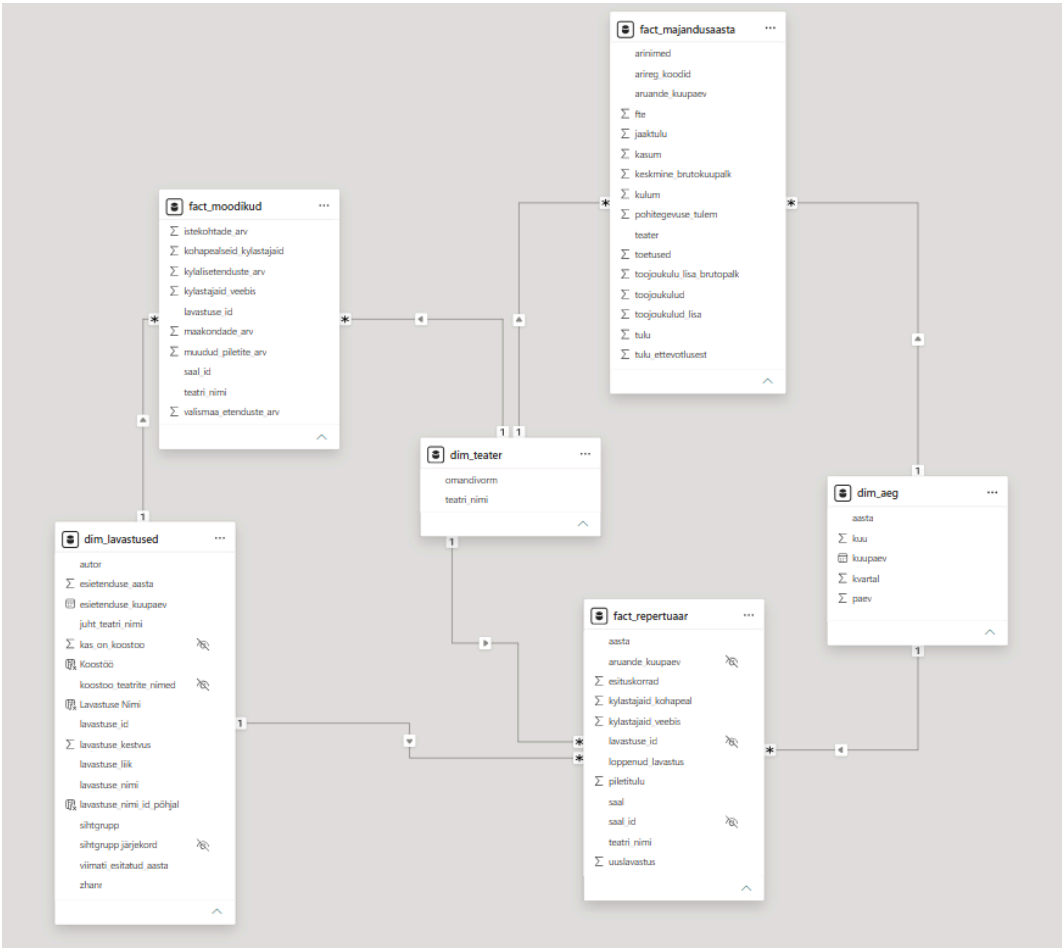
Within PostgreSQL, a structured star schema comprising dimension and fact views was constructed using SQL. Finally, this relational model was imported from PostgreSQL into Power BI for comprehensive analysis and visualization.

### Data Model

For data analysis, a relational data model based on a star schema was created to combine production, theater, repertoire, and statistical data from various sources into a unified analysis environment. The model consists of three dimension tables — `dim_productions`, `dim_theater`, and `dim_time` — and two fact tables: `fact_repertoire` and `fact_measures`.

The modeling followed a core principle: descriptive attributes belong in dimension tables, while analyzable and aggregatable measures belong in fact tables. This structure minimizes data duplication, simplifies relationship management, and enables flexible views and analyses in Power BI segmented by theater, performance, and time period.

### Relational data model based on a star schema



### Dimension tables

**dim\_productions** is the production dimension, designed to describe the productions being analyzed. Each production appears as a unique entry in the model, with `production_id` serving as the primary

identifier. This dimension enables the analysis of productions based on their content features and links production characteristics to performance and attendance indicators stored in the fact tables.

A key task during its creation was ensuring that identical productions across different data sources were treated as a single entity. To achieve this, production names were cleaned and standardized, and secondary attributes like premiere dates were cross-referenced. Additionally, a dedicated attribute was included for co-productions, allowing them to be easily distinguished from single-theater productions.

**dim\_theater** contains descriptive information about theaters, specifically featuring theater\_name and ownership\_type. This dimension links to the fact tables, allowing indicators to be compared by individual theaters and their ownership types. This relationship enables the analysis of production and repertoire statistics grouped by theater, as well as performance comparisons across different ownership structures.

**dim\_time** is a calendar dimension designed to support time-based analysis across various hierarchical levels. It is used to filter and group temporal indicators in the fact tables, enabling trends in repertoire, visitor numbers, and performance metrics to be analyzed across years and periods.

## Fact tables

**fact\_repertoire** is a core fact table containing statistics about repertoire and performances. Its purpose is to link production, theater, and time dimensions while tracking a production's activity within the repertoire.

This table is central to the analysis because it connects performance characteristics with statistical indicators — making it possible to examine, for example, whether productions with a longer lifespan have more showings or visitors, and how these trends change over time.

**fact\_measures** serves as an additional fact table containing measurable indicators related to theaters and halls. It enables the analysis of attendance, performances, ticket sales, and hall metrics at a granular level. Connected to both dim\_productions and dim\_theater, fact\_measures allows these metrics to be evaluated in the context of specific productions and venues.

Within the model, fact tables are not directly connected to one another; instead, cross-fact analysis is handled through shared dimensions. This design effectively prevents data duplication and double counting that can result from directly joining fact tables.

## Using the Data Model in Analysis

The created data model seamlessly links the descriptive characteristics of a production with its actual performances and statistical indicators. For example, a production's lifespan from the dim\_productions table can be connected to performance counts and audience figures in the fact\_repertoire table.

The model enables answers to the project's core research questions, such as:

- how long different productions remain in the repertoire;
- which types of productions have longer or shorter lifespans;
- how the lifespan of productions differs across theaters and ownership forms;
- whether productions with longer lifespans accumulate more performances;
- how a production's lifespan is related to audience numbers;
- what is the relationship between tickets sold and audience attendance;
- how repertoire and attendance have changed over time;

• how the relationship between production and audience changes over time;

- what the share of co-productions is and how they impact statistical indicators.

Granularity:

1. fact\_moodikud: The granularity is at the **theatre, venue/hall, and production level per activity period**. It aggregates general operational metrics and accessibility measures of the theatre.
2. fact\_repertuaar: The granularity is at the **reporting date, theatre, venue/hall, and specific production level**. It enables precise, granular analysis across repertoire and performances.

Data model table volume, structure, and data types

| Table name        | Type            | Number of columns | Number of rows (estimate/volume) | Data types                          | Main fields / columns                                                                                                                                                                                                                                           |
|-------------------|-----------------|-------------------|----------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| `fact_moodikud`   | Fact table      | 10                | ~1 000 – 5 000                   | Integer, Decimal, String/Text       | istekohtade_arv, kohapealseid_kylastajaid, kylalisetenduste_arv, kylastajaid_veebis, lavastuse_id, maakondade_arv, muudud_piletite_arv, saal_id, teatri_nimi, valismaa_etenduste_arv                                                                            |
| `fact_repertuaar` | Fact table      | 11                | ~10 000 – 50 000                 | Integer, Decimal, Date, String/Text | aasta, aruande_kuupaev, esituskorrad, kylastajaid_kohapeal, kylastajaid_veebis, lavastuse_id, loppenuud_lavastus, piletitulu, saal, saal_id, teatri_nimi, uusi_etendusi                                                                                         |
| `dim_lavastused`  | Dimension table | 15                | ~2 000 – 8 000                   | String/Text, Integer, Date          | autor, esietenduse_aasta, esietenduse_kuupaev, H_Lavastus_Silt, juht_teatri_nimi, kas_on_koostoo, koostoo_teatrite_nimed, lavastuse_id, lavastuse_kestvus, lavastuse_liik, lavastuse_nimi, Lavastuse_nimi_id_pohjal, sihtgrupp, viimati_etendatud_aasta, zhantr |
| `dim_teater`      | Dimension table | 2                 | ~30 – 100                        | String/Text                         | omandivorm, teatri_nimi                                                                                                                                                                                                                                         |
| `dim_aeg`         | Dimension table | 5                 | ~4 380 (2014–2025)               | Date, Integer                       | aasta, kuu, kuupaev, kvartal, paev                                                                                                                                                                                                                              |

Data Profile

The primary characteristics analyzed within the dataset included productions, theaters, premiere years, production types, authors' origins, ownership forms, performance counts, visitor numbers, and ticket sales.

The production dataset enabled a thorough examination of repertoire lifecycles and trends. By linking repertoire records with statistical data, it became possible to compare how long productions remained active and evaluate the total performances and audiences they accumulated over time.

However, data profiling also revealed notable limitations. Special attention was directed toward co-productions (KT-productions), where historical inconsistencies and variations in how visitors and performances were reported across different years became apparent. Due to these data inconsistencies, a reliable analysis of KT-productions could not be performed and had to be excluded from the final results. Inconsistencies in the spelling of production and author names were also identified and managed during the data cleaning stages.

## **Data processing**

### **Data Protection**

The data utilized in this project contained no sensitive or personally identifiable information. The analysis relied entirely on publicly available statistics and repertoire data from Estonian theaters, alongside aggregate performance data provided by the Estonian Theatre Agency (ETA). Because the datasets exclusively comprise aggregated metrics regarding theaters, performances, shows, visitors, and economic indicators (such as ticket revenue), additional personal data protection or anonymization measures were not required.

### **Data Quality**

The analysis primarily utilized the "Productions" and "Repertoire" Excel tables as source data, cross-referencing them with public statistical data from the Estonian Theatre Agency / Theatre Union for the period 2014–2025.

Initial data profiling revealed several notable data quality issues.

#### The primary challenges included:

- Variations in the spelling of the same production name across different data tables;
- Inconsistent use of special characters, quotation marks, and other symbols;
- Discrepancies in authors' names, including instances where an author changed their last name over time;
- The association of co-productions (KT-productions) with multiple theaters, resulting in production duplication and inconsistent reporting of visitors and performance counts;
- Inaccurate labeling of theatres' stationary halls. Because specific addresses and hall identifiers (saal\_ID) were frequently absent, hall occupancy rates could not be reliably analyzed.

Consequently, data quality could not be measured simply as a percentage of incorrect or missing rows. Many problems were not technical errors, but structural differences in how data was historically entered and compiled. As a result, standard technical cleaning had to be supplemented with manual verification of the data's actual context and meaning.

### **Data Quality Issues with Collaborative Performances**

The trickiest data quality issue involved handling collaborative performances (KT-productions). A comparison of public repertoire data revealed that between 2014 and 2019, a consistent logic was not applied across all theaters for aggregating visitor and performance counts.

The underlying aggregation rule cannot be clearly deduced from the data: in some instances, visitor numbers for the same collaborative performance differ between participating theaters, while in others they are identical. Although the data has been largely standardized since 2020 (with collaborating theaters reporting identical visitor counts), the lack of a uniform historical rule means that aggregated values cannot be automatically treated as fully comparable across the entire 2014–2025 timeline. Consequently, these data limitations must be kept in mind when interpreting insights related to KT-productions.

Furthermore, handling collaborative productions required care: duplicate entries could not simply be removed. Because a single production associated with multiple theaters often contains crucial context regarding which theater's activities are linked to specific performances and audiences, identifying the production entity and linking it to its respective theaters had to be treated as entirely separate data processing steps.

## **Data Transformation**

During data processing, the primary transformations and data engineering steps included:

- Standardizing text fields and formatting;
- Converting raw date text into structured date formats;
- Correcting the spelling of production titles and author names;
- Merging data tables;
- Identifying and matching identical productions across disparate data sources;
- Flagging collaborative productions (KT-productions);
- Deriving production completion years and lifespan metrics;
- Creating measures needed to calculate the duration or lifespan of a production;
- Integrating public statistics, repertoire records, and production details.

Python was utilized for the initial ETL (Extract, Transform, Load) phase to clean and stage the raw Excel files. Subsequently, a star schema—comprising standardized dimension and fact views—was constructed within a PostgreSQL database using SQL. These database views were then imported into Power BI to establish robust relationships, enabling seamless analysis and visualization of productions, repertoires, performances, and financial metrics within a unified data model.

## **Data Enrichment**

In addition to the original dataset, several derived measures and enriched features were created to enable a deeper comparative analysis of productions. The primary enriched attributes include:

- Last active year: Determined by scanning historical records across all reporting years to identify the year of a production's final recorded performance;
- Production lifespan / duration: Calculated from the premiere date and the last active year to measure the production's total operational active lifetime;
- Collaboration indicator (KP): A specific flag identifying co-productions and collaborative projects.

These enhanced features extend analytical capabilities far beyond simple performance counts, allowing for an in-depth examination of production life cycles, repertoire utilization, and correlations with audience attendance and show frequency.

## Recommendations for Data Flow and Further Data Management

The challenges encountered during this project highlight opportunities to improve the overall quality and consistency of national theatre statistics through standardized data entry and classification practices.

To enhance future reporting, recommendations for the Estonian Theatre Agency include establishing comprehensive data entry guidelines for all participating theatres. These guidelines should specifically define:

- Standardized naming conventions for production titles;
- Uniform formatting for authors' names;
- Clear protocols for identifying and recording co-productions;
- Consistent methodologies for reporting and distributing visitor numbers and performance counts across collaborative projects;
- Criteria for determining when a production constitutes a new entry;
- Guidelines for maintaining unique production identifiers over time;
- Standardized nomenclature and identification for theatre stationary halls.

Furthermore, implementing a dedicated bridge or link table for collaborative productions would be highly beneficial. Such a table would explicitly map productions to all involved theatres while preserving each theatre's specific contribution to performances and attendance. This structural enhancement would prevent the loss of contextual attribution regarding which theatre's activities generated specific audience metrics.

Ultimately, adopting these standardized data management practices would significantly improve cross-year and cross-theatre data comparability and minimize the need for manual interpretation during analysis.

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## Analysis and visualization

Based on the provided project scripts (MAIN\_teatri\_excel\_import.py and MAIN\_view.txt), the data engineering and modeling pipeline executed the following core processing tasks:

### 1. Source Data Ingestion, Cleaning, and Standardization (MAIN\_teatri\_excel\_import.py)

- **File Inclusion:**
  - **Repertoire files:** Excel workbooks for the years 2014–2025 (teatrite\_repertuaar\_{year}.xlsx).
  - **Productions (Lavastused):** Combined workbook 2014–2025LAVastused.xlsx.
  - **Additional tables:** Excel files for production title corrections, theatre business entities, and annual financial statement data.
- **Text and Title Cleaning:**
  - Trimming whitespace and applying Unicode Normalization Form C (NFC).
  - Standardizing quotation marks and apostrophe symbols in production titles, and removing repeated typographical errors.

- Converting column headers to lowercase and replacing spaces and special characters (accented letters, slashes, etc.) with underscores.
  - **Data Type Conversion and Structuring:**
    - Identifying grouping headers in theatre repertoire tables (rows located under category headings without other attribute values) and propagating them across data rows via forward fill.
    - Removing spaces and replacing commas with periods in numeric fields (ticket revenue, visitors, number of performances) to cast them as numeric data types.
    - Appending the column andmeaasta (data year).
- 

## 2. Loading Data into PostgreSQL Database

- **Database Table Creation:**
    - Writing the cleaned staging datasets directly to the PostgreSQL database under the schema teatriliit.
    - Storing data into five dedicated flat tables: repertuaar\_kokku, lavastused\_koond, lavastuse\_nimekorrektsioon, teater\_ariuhing, and majandusaasta\_andmed.
- 

## 3. In-Database Corrections and Creation of SQL Views (MAIN\_view.txt)

- **Database Preprocessing:**
    - **Title corrections:** Updating production titles and premiere dates in the repertoire table based on the title correction reference table.
    - **Venue information:** Enhancing venue details in the repertoire table using data from the combined productions table.
  - **Data Model Building (Dimensions and Fact Tables for Power BI / Reporting):**
    - **dim\_lavastused:** Creates a unique production dimension (with IDs P001, P002, etc.) based on title and premiere date. Identifies the primary lead theatre, co-production partners, and calculates the last year performed as well as production duration in years.
    - **dim\_teater:** Aggregates unique theatres along with their latest known ownership structure.
    - **dim\_saali:** Creates a unique theatre-and-venue key (teatri\_nimi | saali\_nimi) alongside maximum seating capacity.
    - **dim\_aeg:** Generates a daily calendar table for time-series analysis.
    - **fact\_repertuaar:** An annual fact table for analyzing ticket revenues, onsite/online visitor counts, and total number of performances.
    - **fact\_moodikud:** A fact table summarizing total period metrics (2014–2025), such as ticket sales, guest performances, and international shows.
    - **fact\_majandusaasta\_andmed:** An aggregated view of financial indicators from theatre corporate entity annual reports (2019–2025).
- 

## Visualization and Design Analysis

### 1. Theme and Visual Identity (Contextual Design)

Theatre Ticket Concept: The vintage theatre ticket frame and warm beige tone tie dry numbers to the thematic content, making it visually more appealing to follow.

Value of Custom Craftsmanship: The background and details are styled manually and tailored to the layout of each page, ensuring visual integrity and uniqueness. The pages do not feel like mass production or a standard BI template, but rather a intentionally designed product.

2. **Layout, Structure, and Navigation**

Systematic Layout: The dashboards follow BI best practice: key aggregate metrics (KPI cards) are located at the top of the page, while detailed bar and line charts occupy the middle and bottom sections.

Unified Filter System: A consistent filter panel located on the left margin and top corner ensures convenient usability. Filterability (data year, theatre, genre, target group) makes exploring the data intuitive and user-friendly.

Logical Page Structure: The dashboard is divided into topic-based views (e.g., TOP 10, Production Tables, Production Pass, Financial Indicators). This keeps the amount of information on a single page balanced and avoids overwhelming the viewer.

3. **Color Scheme and Visual Hierarchy**

Harmonious Palette: Warm background tones (warm beige, golden) balance contrasting bar charts (dark red, golden yellow, dark blue). Colors help clearly distinguish between different types of metrics (e.g., visitors vs. ticket revenue).

Readability: Data cards (KPIs) are clear and high-contrast, making it easy to quickly grasp key figures.

|                                      |                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Public Dashboard Limitations:</b> | Only aggregated (summary) general data can be shown on a public dashboard: number of performances, overall attendance, types of productions, target audiences, and overall ticket revenue at the theatre level. Individual employees' salary information or the theatre's internal confidential financial forecasts must not be made public. |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# poeti teatri avventura

plans and results

Limitations and Further Developments

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### Limitations (what was not done):

- Production-based cost accounting: Ticket revenue is shown, but detailed cost data is missing (production costs, licenses, work hours spent on the production), so the net profit (ROI) of the production cannot be calculated.
- Theater occupancy rate: Visitor numbers and the number of performances are shown, but the exact theater occupancy percentage (%) cannot be calculated without knowing the actual seating capacity of the theaters

### • How to improve work in the future:

- Profitability model: Integrate the direct costs of each production into the data model to identify the real break-even point.
- Predictive analytics: Create a machine learning model to forecast the revenue and attendance of new productions before the premiere (taking into account the genre, target audience, and premiere timing).
- Adding seasonality and sales channels: Analyze sales dynamics by month and ticket sales channels.